# Chapter 5

# Valuation and Asset Pricing

The preceding chapters built the pieces needed to price a financial asset: Chapter 3 described the institutions and instruments that are traded, and Chapter 4 showed how a firm's financial statements reveal the cash flows it generates. This chapter puts those pieces together and asks the central question of asset pricing: given a stream of future cash flows, or a payoff that depends on some other asset's price, what should that claim be worth today?

The chapter answers it with worked examples for the three main instrument classes introduced in Chapter 3: bonds, equities, and derivatives. Every formula comes with a computation a reader can check by hand or reproduce in the companion notebook, because a practitioner should be able to compute a price, not just describe the logic behind it.

The chapter builds in complexity. It begins with the simplest cash flow structures, a single-maturity bond and a constant-growth dividend stream, where closed-form formulas exist and every number can be verified by hand. It then introduces two ways of estimating the discount rate these formulas depend on, the term structure of interest rates and the cost of capital, before turning to the nonlinear payoffs of options, which require the replication and no-arbitrage arguments of modern derivatives pricing. The chapter can also be read as a single valuation story. Meridian Fabrication, the running example introduced in Chapter 4, first shows how a firm's financing mix determines the discount rate; it is then valued with a market-based multiple and finally from its projected cash flows, and the two values are compared. Bonds, equities, and options are not unrelated topics: the same present-value logic reappears in each.

## 5.1 Discounting and Fixed Income

Every valuation in this book reduces to moving cash across time and charging for the privilege. This part develops that machinery on the instruments where it is cleanest, then asks what happens to a bond's price when the rate used to discount it moves.

### 5.1.1 The Time Value of Money, Revisited

Chapter 1 introduced the present value relation

$$V = \sum_{t=1}^{T} \frac{CF_t}{(1+r)^t}$$

as the unifying idea behind bond pricing, stock valuation, and project appraisal. This chapter makes that idea operational. Two quantities determine the value of any claim on future cash flows: the size and timing of the cash flows themselves, and the discount rate $r$, which compensates the holder for the time value of money and for the risk that the cash flows might not arrive as expected. A safer stream of cash flows is discounted at a lower rate and is therefore worth more, all else equal, than a riskier stream of the same expected size.

Why can the same relation describe such different instruments? A bond's cash flows are contractually fixed (barring default) and known in advance; a stock's cash flows are uncertain and expected to grow; an


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option's payoff is not just uncertain but depends nonlinearly on another asset's future price. What varies across the chapter is not the principle but the complexity of $CF_t$ and the effort required to pin down $r$.

### 5.1.2 Bond Pricing and Yield to Maturity

Bonds come first because their cash flows are the least ambiguous: a fixed schedule of coupons and a principal repayment, specified in the bond's contract at issuance. A bond with face value $F$, annual coupon $C$, and $n$ years to maturity is priced by discounting each promised cash flow at the market yield $y$:

$$P = \sum_{t=1}^{n} \frac{C}{(1+y)^t} + \frac{F}{(1+y)^n}. \tag{5.1}$$

Consider a 3-year bond with face value $F = \$1{,}000$ and an annual coupon rate of 5% (so $C = \$50$ per year), priced at a market yield of $y = 7\%$. Table 5.1 shows each discounted cash flow.

| Year $t$ | Cash flow | Discount factor $(1+y)^{-t}$ | Present value |
|---|---|---|---|
| 1 | \$50.00 | 0.9346 | \$46.73 |
| 2 | \$50.00 | 0.8734 | \$43.67 |
| 3 | \$1,050.00 | 0.8163 | \$857.11 |
| | | **Price** | **\$947.51** |

Table 5.1: Pricing a 3-year, 5% annual-coupon bond at a 7% yield

The bond trades below its \$1,000 face value because the market yield (7%) exceeds the coupon rate (5%); an investor demands compensation via a lower purchase price. The yield to maturity is, conversely, the single discount rate $y$ that makes the present value of the promised cash flows equal to the observed market price; for $n > 1$ this equation generally has no closed-form solution and is instead solved numerically (for example with a root-finding routine such as `scipy.optimize.brentq`).

This inverse relationship between price and yield is one of the most important facts in fixed income and the reason bond portfolios carry interest-rate risk. The sensitivity of price to yield is commonly summarized by duration, a weighted average of the times at which cash flows are received, weighted by the present value of each cash flow. For the bond in Table 5.1, the (Macaulay) duration is approximately 2.86 years, close to but less than the 3-year maturity, because part of the value comes from the earlier coupon payments.

#### Pricing Default Risk

The formula above assumes the promised cash flows arrive with certainty, a reasonable approximation for a government bond but not for a corporate bond, which may default. A simple fix is to discount the *expected* cash flow rather than the promised one, allowing for both the probability of default and the partial recovery a bondholder typically receives after a default.

Consider a simplified one-year, zero-coupon corporate bond with face value \$1,000, a risk-free rate of $r_f = 4\%$, an estimated one-year probability of default $p = 3\%$, and a recovery rate of 40% of face value if default occurs (that is, bondholders recover \$400 per \$1,000 of face value in default). The expected payoff at maturity is

$$\mathbb{E}[\text{Payoff}] = (1-p)(\$1{,}000) + p(0.40)(\$1{,}000) = 0.97(1{,}000) + 0.03(400) = \$982,$$

and discounting this expected payoff at the risk-free rate gives a price of

$$P = \frac{982}{1.04} \approx \$944.23.$$

Equivalently, this can be expressed as a credit spread added to the risk-free rate: for small default probabilities, the required credit spread is approximately

$$p \times (1 - \text{recovery rate}) = 0.03 \times 0.60 = 1.8\%,$$

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giving a corporate bond yield of roughly 4%+1.8% = 5.8% and an approximate price of 1,000/1.058 ≈ \$945.18, close to but not identical to the expected-cash-flow price, since the credit-spread shortcut is itself an approximation. Both approaches price default risk the same way: by widening the discount rate, or equivalently shrinking the expected cash flow, relative to a risk-free bond with the same promised payments. The size of the adjustment depends on the probability of default and the severity of loss, the two quantities Chapter 9's credit models are built to estimate.

### 5.1.3 Interest Rate Risk: Using Duration to Approximate Price Changes

Duration is not just a descriptive statistic; it is the basis of a widely used first-order approximation for how a bond's price responds to a change in yield. Modified duration is defined as

$$D_{\text{mod}} = \frac{D_{\text{Mac}}}{(1+y)},$$

and the approximate percentage price change for a small yield change $\Delta y$ is

$$\frac{\Delta P}{P} \approx -D_{\text{mod}}\,\Delta y. \tag{5.2}$$

For the bond in Table 5.1, whose Macaulay duration of 2.86 years is 2.8553 before rounding, $D_{\text{mod}} = 2.8553/1.07 = 2.6685$ (2.67 to two decimals; the extra digits matter for the dollar figures below). If the market yield rises from 7% to 8% ($\Delta y = +0.01$), the approximation predicts

$$\Delta P \approx -2.6685 \times 0.01 \times \$947.51 \approx -\$25.28, \qquad P_{\text{new}} \approx \$922.23.$$

Repricing the bond exactly at $y = 8\%$ using the original present-value formula gives $P_{\text{new}} = \$922.69$, so the duration approximation is off by about \$0.46, understating the price. If instead the yield falls from 7% to 5% ($\Delta y = -0.02$), the approximation gives $P_{\text{new}} \approx \$998.08$, while the exact price is $P_{\text{new}} = \$1{,}000.00$ (at $y = 5\%$, the yield equals the coupon rate, so the bond prices exactly at par). In both directions the approximation understates the true price, an effect known as convexity: the price-yield relationship curves upward, so a linear approximation always predicts too low a price, whether yields rise or fall. Duration is adequate for small yield changes; for larger moves, or where precision matters, convexity adjustments or full repricing are used.

Convexity can be estimated directly from three prices, the price today and the prices after a small yield increase and decrease, using

$$\text{Convexity} \approx \frac{P_+ + P_- - 2P_0}{P_0(\Delta y)^2},$$

and the second-order (duration-plus-convexity) approximation to the price change is

$$\frac{\Delta P}{P} \approx -D_{\text{mod}}\,\Delta y + \frac{1}{2}\,\text{Convexity}\,(\Delta y)^2.$$

Using $P_0 = \$947.5137$ ($y$=7%), $P_+ = \$922.6871$ ($y$=8%), and $P_- = \$973.2699$ ($y$=6%), carried to four decimals because the numerator is a small difference of large numbers,

$$\text{Convexity} \approx \frac{922.6871 + 973.2699 - 2(947.5137)}{947.5137\,(0.01)^2} = \frac{0.9296}{0.0947514} \approx 9.81.$$

Plugging in the cent-rounded prices \$922.69, \$973.27, and \$947.51 instead gives 9.92: rounding each price by less than half a cent moves the numerator from 0.9296 to 0.94, about one percent, a small-scale warning about numerical second differences that applies equally to the finite-difference Greeks of later chapters. Adding the convexity term to the earlier duration-only forecast for a rise to $y = 8\%$ gives

$$P_{\text{new}} \approx 947.51\left(1 - 2.6685(0.01) + \tfrac{1}{2}(9.81)(0.01)^2\right) \approx \$922.69,$$

which matches the exact repriced value to the penny, against \$922.23 from duration alone. Duration captures the slope of the price-yield relationship and convexity its curvature; using both is standard practice whenever a fixed-income portfolio's risk is measured precisely, for example in the value-at-risk calculations of Chapter 8.

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**Portfolio Duration**

Duration extends naturally from a single bond to a portfolio of bonds: the portfolio's duration is the market-value-weighted average of the durations of its individual holdings,

$$D_{\text{portfolio}} = \sum_i w_i D_i,$$

where $w_i$ is the fraction of portfolio market value invested in bond $i$, the same weighted-average structure used for portfolio return in Chapter 2 and for portfolio variance in Chapter 12. Suppose a portfolio holds 60% of its market value in the 3-year bond from Table 5.1 and 40% in a 5-year zero-coupon bond priced at a 7% yield. A zero-coupon bond has a single cash flow, so its duration equals its maturity, 5 years. Using the coupon bond's full-precision duration of 2.8553 years rather than the rounded 2.86, the portfolio duration is

$$D_{\text{portfolio}} = 0.6(2.8553) + 0.4(5.0) \approx 3.71 \text{ years}.$$

This single number lets a portfolio manager approximate the effect of a parallel rate shift on the whole portfolio with the modified-duration approximation, without repricing each bond, which matters for portfolios with hundreds of positions. Managing duration, for example by shifting the mix of short- and long-maturity bonds toward a target, is one of the most basic fixed-income risk management techniques, and it reappears in the market risk measures of Chapter 8.

### 5.1.4 The Term Structure of Interest Rates

Every bond example so far has used a single discount rate $y$ for all maturities, but the market yield on a bond typically depends on its maturity, a relationship called the term structure of interest rates, or the yield curve. Table 5.2 shows a stylized, upward-sloping yield curve for government bonds of different maturities.

| Maturity | 1 year | 2 years | 5 years | 10 years | 30 years |
|---|---|---|---|---|---|
| Yield | 4.2% | 4.4% | 4.8% | 5.2% | 5.5% |

Table 5.2: A stylized government bond yield curve

An upward-sloping curve like the one in Table 5.2 is the most common shape historically and is usually attributed to two effects: investors demand a higher yield for the greater interest-rate risk of a longer-maturity bond (a term premium), and the market may expect short-term rates to rise, which raises long-term yields today. When the curve inverts, with short-term yields above long-term yields, the market is often read as expecting rate cuts, typically because it anticipates an economic slowdown. That is why inversions are watched as a potential recession indicator, the kind of macroeconomic time-series relationship studied in Chapter 7.

Once a full yield curve is available, each cash flow should in principle be discounted at the spot rate for its own maturity, not at the bond's yield to maturity, which is a complex weighted average of those spot rates. The distinction matters most for bonds with cash flows spread over a long horizon and is central to pricing and hedging any rate-sensitive instrument precisely; the single-rate approximation used in the rest of this chapter is adequate for building intuition.

**Worked Example 5.1: Extracting Forward Rates from the Spot Curve**

The spot rates in Table 5.2 apply to borrowing or lending that starts today. A forward rate is the rate, implied by today's spot curve, for a loan that starts at a future date and ends at a later one, and no-arbitrage pins it down by the same replication logic as Chapter 3's no-arbitrage bond example. Investing for two years at the 2-year spot rate must pay the same as investing for one year at the 1-year spot rate and then reinvesting for a second year at the rate that can be locked in today for that year,

$$(1+y_2)^2 = (1+y_1)(1+f_{1,2}) \quad \Longrightarrow \quad f_{1,2} = \frac{(1+y_2)^2}{1+y_1} - 1. \tag{5.3}$$

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Using Table 5.2's 1-year rate of 4.2% and 2-year rate of 4.4%,

$$f_{1,2} = \frac{1.044^2}{1.042} - 1 \approx 4.60\%,$$

higher than both the 1-year and 2-year spot rates, as an upward-sloping curve implies: the rate for the second year alone must exceed the 2-year average to pull that average up from the lower 1-year rate. The same logic extends to any pair of maturities; using the 2-year and 5-year spot rates, the implied 3-year forward rate spanning years 2 through 5 is

$$f_{2,5} = \left(\frac{(1+y_5)^5}{(1+y_2)^2}\right)^{1/3} - 1 \approx 5.07\%,$$

again above both the 2-year and 5-year spot rates, consistent with the curve's continued upward slope. Forward rates matter beyond the arithmetic. They are the rates a company can lock in today with a forward-starting swap or a forward-rate agreement (relatives of the swaps and forwards of Chapter 3) to hedge future borrowing costs. Under the pure expectations theory of the term structure they can also be read, subject to the term-premium caveat above, as the market's current forecast of future short-term rates.

## 5.2 Valuing Equity

An equity claim has no maturity and no promised payment, so its value depends entirely on what is expected of it and on the rate demanded in exchange. This part covers where that rate comes from and the two families of method built on it.

### 5.2.1 The Cost of Capital: CAPM and WACC

Every valuation formula in this chapter takes a discount rate $r$ as an input, and the examples so far, like the dividend discount examples that follow, have simply been handed one. This section shows how that rate is estimated. Firms are financed by a mix of debt and equity, and each source of capital has its own cost.

The cost of debt is the most straightforward: the yield the firm currently pays on its debt (or would pay on new debt of similar risk), adjusted for the tax deductibility of interest payments, since interest expense reduces taxable income,

$$r_d^{\text{after-tax}} = r_d^{\text{pretax}}(1-\tau),$$

where $\tau$ is the firm's marginal tax rate. The cost of equity is harder to observe directly, since equity has no contractual payment, and is typically estimated using the Capital Asset Pricing Model (CAPM), covered in full in Chapter 12,

$$r_e = r_f + \beta(r_m - r_f), \tag{5.4}$$

where $r_f$ is the risk-free rate, $r_m - r_f$ is the market risk premium (the extra return investors demand for holding the overall market rather than a risk-free asset), and $\beta$ measures the stock's sensitivity to market-wide movements. The weighted average cost of capital (WACC) then blends the two financing costs in proportion to their share of the firm's total capital,

$$\text{WACC} = \frac{E}{E+D}\, r_e + \frac{D}{E+D}\, r_d^{\text{after-tax}}, \tag{5.5}$$

where $E$ and $D$ are the market (or, in the absence of market prices, book) values of equity and debt.

Using Meridian Fabrication's Year 2 figures from Chapter 4,

$$E = \$54.0 \text{ million (equity)}, \qquad D = \$58.0 \text{ million (total debt)},$$
$$r_f = 4\%, \qquad \text{Market risk premium} = 6\%,$$
$$\beta = 1.2, \qquad \text{Pretax cost of debt} = 4.0/58.0 \approx 6.90\%,$$
$$\text{Tax rate} = 25\%,$$

where the pretax cost of debt is read off Meridian's own statements rather than assumed: Year 2 interest expense of \$4.0 million on the \$58.0 million of debt outstanding at year end. A rate nearly three percentage

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points above the risk-free rate is what one would expect for a firm whose Altman Z′-score places it in the gray zone (Chapter 4). Then

$$r_e = 0.04 + 1.2(0.06) = 11.2\%, \qquad r_d^{\text{after-tax}} = 0.0690(1 - 0.25) \approx 5.17\%,$$

$$\text{WACC} = \frac{54.0}{112.0}(11.2\%) + \frac{58.0}{112.0}(5.17\%) \approx 0.482(11.2\%) + 0.518(5.17\%) \approx 8.08\%.$$

Book equity stands in for $E$ because Meridian's shares have no market price in this example; Section 5.4 returns to what changes when a market-based equity value is used instead. This WACC of 8.08%, close to the 8% assumed in Chapter 4's economic-profit example, is the discount rate for Meridian's free-cash-flow valuation and the one that belongs in the economic-profit formula, Invested capital × (ROIC − WACC). A firm earning a return on invested capital below its WACC of 8.08% destroys value even while reporting positive accounting profit. WACC is also sensitive to its inputs: a beta of 1.5 instead of 1.2 would raise the cost of equity to $0.04 + 1.5(0.06) = 13\%$ and the WACC to roughly 8.95%, the discount-rate uncertainty flagged as a challenge later in this chapter.

### 5.2.2 The Dividend Discount Model for Equities

Equity valuation follows the same discounting logic but applies it to an infinite, growing stream of dividends rather than a fixed set of coupons. The Gordon growth model prices a share as

$$P_0 = \frac{D_1}{r - g}, \tag{5.6}$$

where $D_1$ is the dividend expected next year, $r$ is the required rate of return on equity, and $g < r$ is the assumed constant long-run growth rate of dividends. For example, if a firm is expected to pay a dividend of $D_1 = \$3.00$ next year, investors require a return of $r = 9\%$, and dividends are expected to grow at $g = 4\%$ per year indefinitely, the model implies a fair price of

$$P_0 = \frac{3.00}{0.09 - 0.04} = \frac{3.00}{0.05} = \$60.00.$$

The model's simplicity is also its main weakness: the price is extremely sensitive to the gap $r - g$, so small changes in the assumed growth or discount rate produce large changes in value. That sensitivity is the practical lesson: a valuation built on this model should be stress-tested over a range of plausible $(r, g)$ pairs, not reported as a single point estimate.

A constant growth rate forever is a strong assumption, since fast-growing firms rarely sustain high growth indefinitely. The two-stage dividend discount model relaxes this by assuming a high growth rate $g_1$ for the first $n$ years, followed by a lower, sustainable growth rate $g_2 < r$ thereafter. The price is the present value of the high-growth dividends plus the present value of a terminal value computed with the Gordon growth formula at the end of the high-growth period:

$$P_0 = \sum_{t=1}^{n} \frac{D_0(1 + g_1)^t}{(1 + r)^t} + \frac{1}{(1 + r)^n}\left[\frac{D_0(1 + g_1)^n(1 + g_2)}{r - g_2}\right].$$

For example, suppose a firm's most recent dividend was $D_0 = \$3.00$, it is expected to grow at $g_1 = 12\%$ for the next 3 years before settling into a stable long-run growth rate of $g_2 = 4\%$, and investors require $r = 9\%$. The high-growth dividends are

$$D_1 = \$3.36, \qquad D_2 = \$3.76, \qquad D_3 = \$4.21,$$

with a combined present value of \$9.50. The terminal value at the end of year 3 is

$$\frac{D_3(1 + g_2)}{r - g_2} = 4.21(1.04)/0.05 \approx \$87.67,$$

which has a present value of \$67.70. Adding the two pieces gives

$$P_0 \approx \$9.50 + \$67.70 = \$77.20.$$

Notice that the terminal value, representing everything beyond year 3, accounts for roughly 88% of the total price; this is typical of growth-stage valuations and is why the assumed terminal growth rate $g_2$, not the near-term high growth rate, is usually the most consequential assumption in the model.

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**Preferred Stock: A Perpetuity Special Case**

Preferred stock typically pays a fixed dividend indefinitely, with no growth at all, which makes the Gordon growth formula collapse to the special case $g = 0$:

$$P_0 = \frac{D}{r}.$$

This is a growing perpetuity with zero growth, sometimes called a level perpetuity. For example, a preferred share paying a fixed annual dividend of $D = \$4.50$, with investors requiring $r = 7.5\%$ given the security's risk, is worth $P_0 = 4.50/0.075 = \$60.00$. Because its dividends are fixed, preferred stock is valued more like a bond than like common stock, even though it is legally equity: for valuation, an instrument's economic behavior, here a fixed, bond-like cash flow, matters more than its legal label.

### 5.2.3 Relative Valuation: Multiples

Every method so far is an absolute valuation method: it estimates intrinsic value from a model of the security's cash flows. Relative valuation instead prices an asset by how similar assets are currently priced, using a valuation multiple.

The most common multiple for valuing an entire business is enterprise value to EBITDA (earnings before interest, taxes, depreciation, and amortization), since EBITDA approximates operating cash flow before financing and accounting choices and enterprise value (the value of both debt and equity claims combined) is capital-structure-neutral, unlike equity value alone. The method has three steps: compute the target firm's EBITDA, multiply it by a multiple observed for comparable, publicly traded firms, and then back out the implied equity value by subtracting net debt (total debt minus cash).

Applying this to Meridian Fabrication: EBITDA is EBIT plus depreciation and amortization, $16.0 + 8.0 = \$24.0$ million. Suppose comparable industrial parts manufacturers currently trade at an EV/EBITDA multiple of $6.5\times$. The implied enterprise value is

$$EV = \text{EBITDA} \times \text{Multiple} = 24.0 \times 6.5 = \$156.0 \text{ million}.$$

Net debt is total debt minus cash, $58.0 - 9.0 = \$49.0$ million, so the implied equity value is

$$\text{Equity value} = EV - \text{Net debt} = 156.0 - 49.0 = \$107.0 \text{ million}.$$

This is nearly double Meridian's \$54.0 million book equity from Chapter 4's balance sheet, and the two need not be close: book equity is an accounting construct built from historical costs, while market-implied equity value is a forward-looking assessment of what the business is worth.

Relative valuation is fast and grounded in observable prices, but it depends entirely on the comparable multiple. If the comparables are overvalued or undervalued, cyclically distorted, or not comparable in growth, margins, and risk, the valuation inherits the error. Absolute (discounted-cash-flow) and relative valuation are best used together: a DCF built on explicit assumptions about growth and discount rates, checked against a multiple grounded in current market pricing. A large disagreement between them is itself information, since it forces the analyst to ask whose assumptions, the DCF's or the market's, are more likely to be right.

The table below summarizes the chapter's valuation tools, including the option-pricing methods developed next, and when each is most useful.

## 5.3 Option Pricing

An option's payoff is not linear in the underlying price, which makes discounting an expected value the wrong tool. This part builds the replication argument that replaces it, works it up to Black-Scholes-Merton, and follows the consequences into risk measurement and corporate decisions.

### 5.3.1 No-Arbitrage Pricing and the Binomial Option Model

Every valuation so far has followed the same recipe: find the cash flow (or expected cash flow) an instrument delivers, then discount it at a rate reflecting its risk. An option seems as if it should work the same way.

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| Method | Best used for | Main inputs | Main caveat |
|---|---|---|---|
| Bond pricing | Fixed-income claims with known coupons | Coupon, maturity, face value, yield | Assumes the promised cash flows are paid as stated |
| Dividend discount model | Mature, dividend-paying equities | Dividend, growth, required return | Very sensitive to the gap $r - g$ and the terminal growth assumption |
| WACC-based DCF | Firms or projects with a mix of debt and equity | Free cash flow, growth, cost of debt, cost of equity, capital structure | Discount-rate uncertainty can dominate the final value |
| Multiples | Quick relative valuation | EBITDA or earnings, peer multiple, net debt | Depends on the comparables being truly similar |
| Binomial or Black-Scholes-Merton | Options and other nonlinear payoffs | Underlying price, strike, volatility, time, rates | Assumes simplifying conditions such as constant volatility |

Table 5.3: A quick guide to the main valuation tools in this chapter

It does not, for a structural reason. A bond's coupon and a stock's expected dividend are each a single number to discount; an option's payoff depends on where the underlying price ends up, and that dependence is not a straight line. A European call option with strike $K$ pays $\max(S_T - K, 0)$ at maturity $T$, where $S_T$ is the underlying asset price at maturity; a European put pays $\max(K - S_T, 0)$.

Figure 5.1 plots both payoffs as a function of the terminal stock price $S_T$ for a strike of $K = \$50$. The call is worthless ("out of the money") for any $S_T \leq K$ and rises one-for-one with $S_T$ once $S_T$ exceeds $K$ ("in the money"); the put shows the mirror-image pattern. This kink at $S_T = K$ makes the payoff nonlinear and calls for the replication argument below instead of a discounted-cash-flow calculation.

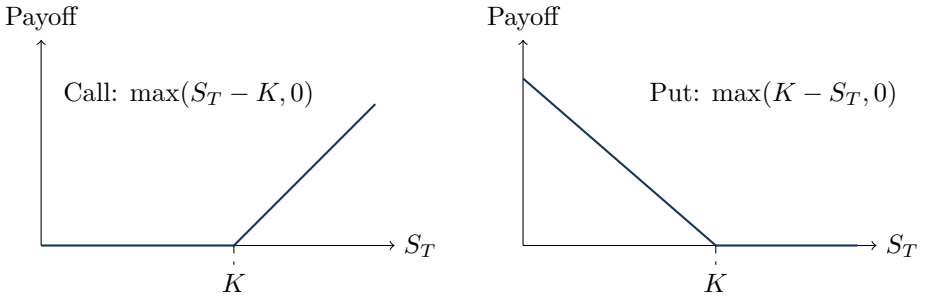



Figure 5.1: Payoff at maturity for a call option (left) and a put option (right) with strike $K$

Options are therefore priced by constructing a replicating portfolio of the underlying asset and a risk-free bond that produces the same payoff in every future state, and then invoking the no-arbitrage principle of Chapter 3: two portfolios with identical payoffs must have the same price today.

The simplest version of this argument is the one-step binomial model. Suppose a stock currently trades at $S_0 = \$50$ and, over one period, either rises to $S_0 u$ or falls to $S_0 d$, with $u = 1.2$ and $d = 0.9$. A risk-free investment over the same period grows by a gross factor $R = 1 + r_f = 1.04$. Consider a call option struck at $K = \$50$. The stock will be worth $S_u = 60$ or $S_d = 45$, so the call will be worth $C_u = \max(60 - 50, 0) = \$10$ or $C_d = \max(45 - 50, 0) = \$0$. No-arbitrage pricing does not need the stock's real-world probability of rising or falling. It asks instead what probability of an up-move makes a portfolio of the stock and a risk-free bond,

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chosen to replicate the option's payoff in both states, price correctly today. That probability, not a forecast, values the option as a discounted expectation under the risk-neutral probability

$$q = \frac{R-d}{u-d} = \frac{1.04-0.9}{1.2-0.9} = 0.467, \tag{5.7}$$

so that

$$C_0 = \frac{1}{R}[q\,C_u + (1-q)\,C_d] = \frac{1}{1.04}[0.467(10) + 0.533(0)] \approx \$4.49. \tag{5.8}$$

The probability $q$ is not the investor's real-world belief about the probability of an up-move; it is the probability implied by the no-arbitrage condition, which is why this is called risk-neutral pricing. The same logic extends, by adding more steps and shrinking the time interval, to the continuous-time model presented next.

**Extending to Two Steps**

A single step is enough to introduce the mechanics, but a multi-step tree whose up- and down-moves are calibrated to the underlying's volatility is what converges to the Black-Scholes-Merton price, as the companion notebook shows for trees of up to 5,000 steps. Extending the same stock and option to two periods, each still with $u = 1.2$, $d = 0.9$, $R = 1.04$, produces three possible terminal prices: $S_{uu} = 50(1.2)^2 = \$72$, $S_{ud} = 50(1.2)(0.9) = \$54$, and $S_{dd} = 50(0.9)^2 = \$40.50$, with call payoffs $C_{uu} = \$22$, $C_{ud} = \$4$, and $C_{dd} = \$0$. The risk-neutral probability $q = 0.467$ is unchanged, since it depends only on $u$, $d$, and $R$, not on the number of steps, so the tree is solved by working backward one period at a time:

$$C_u = \frac{1}{R}[q\,C_{uu} + (1-q)\,C_{ud}] = \frac{1}{1.04}[0.467(22) + 0.533(4)] \approx \$11.92, \qquad C_d = \frac{1}{R}[q\,C_{ud} + (1-q)\,C_{dd}] \approx \$1.79,$$

$$C_0 = \frac{1}{R}[q\,C_u + (1-q)\,C_d] \approx \$6.27.$$

Neither this two-step price of \$6.27 nor the one-step \$4.49 should be compared with the Black-Scholes-Merton price of \$6.88 computed later: the hand tree's $u$ and $d$ imply a lower volatility than the 30% the Black-Scholes-Merton example uses, so moving from one step to two with $u$ and $d$ fixed illustrates backward induction, not convergence. Convergence requires each step's moves to be calibrated to the same annualized volatility.

The $u = 1.2$ and $d = 0.9$ above were chosen for hand computation. A tree with any number of steps instead derives $u$ and $d$ from the annualized volatility using the Cox-Ross-Rubinstein parametrization $u = e^{\sigma\sqrt{\Delta t}}$, $d = 1/u$, where $\Delta t$ is the length of one step, so every tree reflects the same annualized volatility $\sigma$. Table 5.4 prices the same at-the-money call ($S_0 = K = \$50$, $r = 4\%$, $\sigma = 30\%$, $T = 1$ year) with CRR trees of increasing size; the price oscillates as it converges, and the error shrinks from \$1.40 at one step to under \$0.001 by 5,000 steps, confirming that Black-Scholes-Merton is the limit of the discrete tree as the step size shrinks to zero.

| Steps | Binomial price | Error vs. Black-Scholes-Merton |
|---:|---:|---:|
| 1 | \$8.28 | +\$1.40 |
| 2 | \$6.21 | −\$0.67 |
| 5 | \$7.16 | +\$0.28 |
| 20 | \$6.80 | −\$0.07 |
| 100 | \$6.86 | −\$0.01 |
| 500 | \$6.874 | −\$0.003 |
| 1,000 | \$6.875 | −\$0.001 |
| 5,000 | \$6.876 | −\$0.0003 |
| $\infty$ (Black-Scholes-Merton) | \$6.8766 | — |

Table 5.4: Convergence of the CRR binomial call price to the Black-Scholes-Merton price as the number of steps increases ($S_0 = K = \$50$, $r = 4\%$, $\sigma = 30\%$, $T = 1$ year)

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Figure 5.2 lays out the full tree, alongside its American put counterpart introduced below, side by side: stock prices grow forward from $S_0$ on the left to three recombining terminal nodes on the right, while option values are computed in the opposite direction, starting from the known terminal payoffs and working backward one period at a time using $q$.

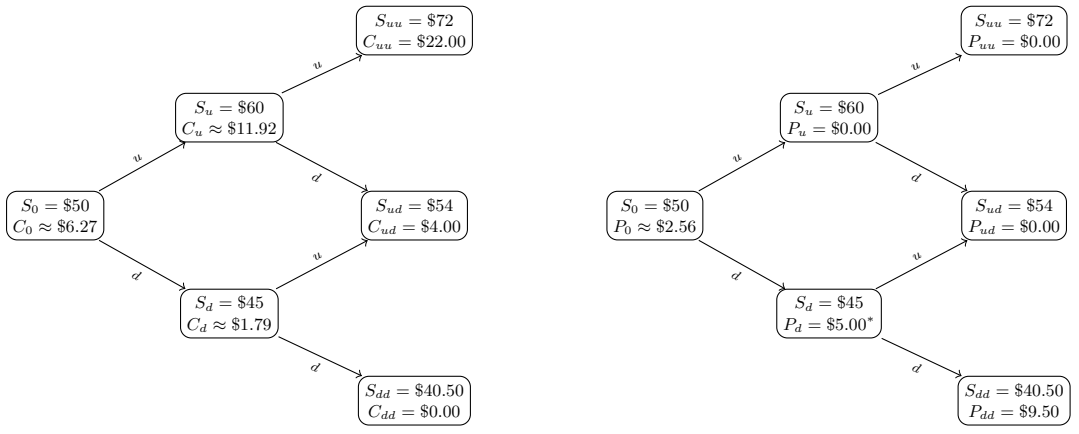
### 5.3.2 American Options and Early Exercise

Every option considered so far has been European, exercisable only at maturity. An American option can be exercised at any time up to and including maturity, and the binomial tree extends naturally to price this feature: at each node, compare the value of exercising immediately (the intrinsic value) to the value of continuing to hold the option, and take whichever is larger.

Consider an American put with the same two-period tree and strike $K = \$50$. At maturity the payoffs are unchanged: $P_{uu} = \$0$, $P_{ud} = \$0$, $P_{dd} = \max(50 - 40.50, 0) = \$9.50$. Working backward, the European continuation value at the down node ($S_d = \$45$ after one down move) is $P_d^{\text{cont}} = [q(0) + (1-q)(9.50)]/1.04 \approx \$4.87$. But immediate exercise at that node is worth $\max(50 - 45, 0) = \$5.00$, which exceeds the continuation value, so an optimal holder would exercise early rather than wait: $P_d^{\text{American}} = \max(5.00, 4.87) = \$5.00$. At the up node, both exercise and continuation are worth \$0 (the put is out of the money after an up move), so $P_u^{\text{American}} = \$0$. Rolling back to today, the continuation value is $[q(0) + (1-q)(5.00)]/1.04 \approx \$2.56$, and since immediate exercise today is worth $\max(50 - 50, 0) = \$0$, the American put is worth $P_0^{\text{American}} \approx \$2.56$.

By contrast, the European put (never allowed to exercise early) is worth only $P_0^{\text{European}} \approx \$2.50$, computed by discounting the same terminal payoffs without ever checking intrinsic value along the way. The American put's extra \$0.07 of value (the full-precision values are \$2.5641 and \$2.4984; subtracting the cent-rounded prices would suggest \$0.06) is the early-exercise premium, and it exists because early exercise was optimal at the down node. This feature is why American puts (and American calls on dividend-paying underlyings) generally have no closed-form formula like Black-Scholes-Merton and require a tree, a finite-difference method, or a simulation that can check the exercise decision at every point in time.

In Figure 5.2 the two trees share identical stock prices, since $u$, $d$, and $q$ do not depend on the option being priced; only the option values on the bottom line of each node differ. The asterisk at the American put's down node marks where the logic diverges: every other node is a discounted expectation of its children, but $P_d = \$5.00$ replaces a \$4.87 continuation value because immediate exercise is worth more. That single substitution is the entire early-exercise premium.




European call American put

Figure 5.2: The two-step binomial tree for the \$50-strike European call (left) and American put (right). Both share the same stock-price tree, growing forward from $S_0$ under up- and down-moves $u = 1.2$ and $d = 0.9$ and recombining at the middle terminal node. Option values (bottom line of each node) are computed backward from the known terminal payoffs using the risk-neutral probability $q = 0.467$, except at the American put's down node, marked with an asterisk, where the \$5.00 intrinsic value of immediate exercise replaces the lower \$4.87 continuation value, producing the early-exercise premium discussed in the text.

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### 5.3.3 The Black-Scholes-Merton Model

The Black-Scholes-Merton model is the continuous-time limit of the binomial approach and remains the benchmark option pricing formula in practice and theory. For a non-dividend-paying stock with current price $S_0$, strike $K$, risk-free rate $r$ (continuously compounded), volatility $\sigma$, and time to maturity $T$ (in years), the price of a European call is

$$C = S_0 N(d_1) - K e^{-rT} N(d_2), \qquad d_1 = \frac{\ln(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T}, \tag{5.9}$$

where $N(\cdot)$ is the standard normal cumulative distribution function. Using the same at-the-money strike as the binomial example, $S_0 = K = \$50$, with $r = 4\%$, $\sigma = 30\%$, and $T = 1$ year,

$$d_1 = \frac{0 + (0.04 + 0.045)(1)}{0.30} = 0.283, \qquad d_2 = 0.283 - 0.30 = -0.017,$$

and looking up $N(0.283) = 0.6115$ and $N(-0.017) = 0.4934$ gives

$$C = 50(0.6115) - 50\, e^{-0.04}(0.4934) \approx \$6.88.$$

This is the price the CRR tree of Table 5.4, calibrated to the same $\sigma = 30\%$, approaches as its step size shrinks, not the hand-computed \$4.49 and \$6.27, whose $u = 1.2$ and $d = 0.9$ imply a lower volatility. The corresponding put price is

$$P = K e^{-rT} N(-d_2) - S_0 N(-d_1) \approx \$4.92,$$

and the two prices must satisfy put-call parity,

$$C + K e^{-rT} = P + S_0, \tag{5.10}$$

which can be checked directly: $6.88 + 50\, e^{-0.04} = 6.88 + 48.04 = \$54.92$, and $4.92 + 50 = \$54.92$, confirming consistency. Put-call parity holds regardless of any particular pricing model, since it follows from no-arbitrage alone; it is therefore a useful sanity check on any option pricing implementation, including a machine learning model trained to approximate option prices.

The quantity $N(d_1) = 0.6115$ in this example is also the option's delta, the sensitivity of the option price to a one-dollar change in the stock price, and is the first of several "Greeks" used to measure and hedge an option position's risk exposures.

### 5.3.4 The Greeks: Measuring an Option's Risk Exposures

A market maker who has just sold the call option priced above is exposed to several distinct sources of risk, and each of the Greeks quantifies one of them. Four of the five Greeks below share a single building block: $\phi(\cdot)$, the standard normal probability density function,

$$\phi(x) = \frac{1}{\sqrt{2\pi}}\, e^{-x^2/2},$$

the height of the standard normal bell curve at $x$, distinct from $N(x)$, the cumulative area under that curve already used to price the option itself. Evaluated at the same $d_1$ computed above, but at full precision ($d_1 = 0.28333$, not the rounded 0.283 displayed there, since rounding before this step would cascade into every Greek that uses it),

$$\phi(d_1) = \frac{1}{\sqrt{2\pi}}\, e^{-0.28333^2/2} \approx 0.3832.$$

For a European call under Black-Scholes-Merton,

$$\Delta = N(d_1), \qquad \text{(sensitivity to the stock price)}$$

$$\Gamma = \frac{\phi(d_1)}{S_0\sigma\sqrt{T}}, \qquad \text{(sensitivity of delta itself to the stock price)}$$

$$\text{Vega} = S_0\phi(d_1)\sqrt{T}, \qquad \text{(sensitivity to volatility)}$$

$$\Theta = -\frac{S_0\phi(d_1)\sigma}{2\sqrt{T}} - rKe^{-rT}N(d_2), \qquad \text{(sensitivity to the passage of time)}$$

$$\rho = KTe^{-rT}N(d_2). \qquad \text{(sensitivity to the interest rate)}$$

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Evaluating each at the same inputs as before ($S_0 = K = \$50$, $r = 4\%$, $\sigma = 30\%$, $T = 1$) gives $\Delta = 0.6115$ (already computed), $\Gamma = \phi(d_1)/(50 \times 0.30 \times 1) \approx 0.0255$, Vega $= 50 \times \phi(d_1) \times 1 \approx 19.16$, $\Theta \approx -3.82$ per year (about $-\$0.0105$ per calendar day), and $\rho = 23.70$ (per unit, i.e., per 100 percentage points, change in the interest rate). The shared $\phi(d_1) \approx 0.3832$ term is why gamma, vega, and theta rise and fall together. Whatever pushes $d_1$ toward zero, the peak of the bell curve (an at-the-money option, or higher volatility relative to the stock's distance from the strike), raises gamma, vega, and the magnitude of theta at once. Delta and rho depend only on $N(d_1)$ and $N(d_2)$ and do not share this driver.

These numbers have direct trading interpretations. Delta of 0.6115 means the option's price moves about \$0.61 for every \$1 move in the stock, which is why a market maker who sold this call would typically hedge by buying 0.6115 shares per option sold, a strategy called delta hedging. Gamma of 0.0255 measures how quickly that hedge ratio itself changes as the stock moves, so a market maker must rebalance the hedge more frequently when gamma is high. Vega of 19.16 (often quoted per percentage point of volatility, $19.16/100 \approx \$0.19$) shows the option gains or loses value from changes in volatility expectations alone, independent of the stock price; this is the channel through which the volatility clustering of Chapter 7 affects option values. Theta of $-\$3.82$ per year reflects time decay: all else fixed, the option loses value as expiration approaches, since less time remains for the stock to move in the holder's favor.

**Worked Example 5.2: Delta Hedging and Its Limits**

Suppose a market maker sells 1,000 of these call options and immediately delta-hedges by buying $1{,}000 \times 0.6115 = 611.5$ shares. If the stock rises by \$1, from \$50 to \$51, the hedge's stock position gains $611.5 \times \$1 = \$611.50$. Recomputing the exact Black-Scholes-Merton price at $S = \$51$ (holding every other input fixed) gives a new call price of \$7.500779, up from \$6.876632, so the short option position loses $1{,}000 \times (7.500779 - 6.876632) = \$624.15$ (the four-decimal prices \$7.5008 and \$6.8766 would give \$624.20, a rounding artifact). The hedge does not offset this loss exactly: the net result is $611.50 - 624.15 = -\$12.65$, small relative to the \$1,000-contract position but not zero, because delta is only a first-order (linear) approximation of the option's response to the stock price, the same limitation as the modified-duration approximation for bonds. Just as convexity corrected the bond-duration approximation, gamma corrects the delta approximation here: the delta-only prediction for the new option price is $C_0 + \Delta(S_1 - S_0) = 6.8766 + 0.6115(1) = \$7.4882$, while adding the gamma correction, $C_0 + \Delta(S_1 - S_0) + \frac{1}{2}\Gamma(S_1 - S_0)^2 = 7.4882 + 0.5(0.0255)(1)^2 \approx \$7.5009$, lands almost exactly on the true recomputed price of \$7.5008.

A market maker who rebalances only occasionally is therefore exposed to residual gamma risk between rebalances. That risk grows with the size of the stock move (the gamma term grows with its square) and is largest for at-the-money options near expiration, which have the highest gamma. That is why options market makers manage gamma as well as delta, and why one large, sudden move costs a delta-hedged book more than a series of small moves of the same total size.

### 5.3.5 Implied Volatility

The Black-Scholes-Merton formula takes $\sigma$ as an input and produces a price as output. Market participants often run it in reverse: given an option's market price, they solve for the volatility that reproduces that price, called the implied volatility. Because the formula has no algebraic inverse, implied volatility is found numerically, with the same kind of root-finding used for yield to maturity in Section 5.1.2.

Suppose the call priced at \$6.88 under an assumed volatility of 30% is instead observed trading at \$7.50. The implied volatility $\hat{\sigma}$ solves $C_{BS}(\hat{\sigma}) = 7.50$, which a numerical solver finds to be $\hat{\sigma} \approx 33.25\%$, above the 30% assumed. The market is pricing in more uncertainty than the 30% historical-volatility assumption suggests, perhaps because of an upcoming earnings announcement, macroeconomic uncertainty, or a different view. Because implied volatility is forward-looking, reflecting current expectations rather than a historical average, it is often preferred to historical volatility for pricing related options.

If the Black-Scholes-Merton model's constant-volatility assumption held exactly, every option on the same underlying and maturity would imply the same $\hat{\sigma}$ whatever its strike: calls at $K = \$40$ and $K = \$60$ would back out the same number, because the model treats $\sigma$ as a property of the stock, not of the contract. In practice, implied volatility plotted against strike traces a curve, not a flat line, in one of two common shapes. A **volatility smile** is roughly symmetric: implied volatility is lowest at the money and rises toward both deep in-the-money and deep out-of-the-money strikes, a pattern most associated with currency options,

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where a large move in either direction is plausible. A **volatility skew**, the shape observed for equity and equity-index options, slopes mostly downward: implied volatility is highest for low strikes (deep in-the-money calls, equivalently far out-of-the-money puts) and falls as the strike rises, so out-of-the-money puts are priced as if the stock were more volatile than at-the-money or out-of-the-money calls suggest.

Two economic explanations are offered for the equity skew, and both remain subjects of empirical research rather than settled fact. The first is the *leverage effect* (Black 1976): a falling stock price raises a levered firm's debt-to-equity ratio and therefore its equity volatility, so low-stock-price states are more volatile ones. The second is *crash risk*, sometimes called "crashophobia" (Rubinstein 1994): the pronounced equity skew now taken for granted was far less visible in listed U.S. index option prices before the October 1987 crash, and persistent demand for downside protection, out-of-the-money puts bought as portfolio insurance, has kept those puts priced at a premium (a higher implied volatility) ever since.

### 5.3.6 Worked Example 5.3: Computing an Implied Volatility Skew

The single implied-volatility computation above generalizes directly: solving $C_{BS}(\hat{\sigma}) =$ market price separately at each strike, holding $S_0 = \$50$, $r = 4\%$, and $T = 1$ year fixed throughout, traces out the skew itself. Table 5.5 does this for five hypothetical market prices spanning strikes of \$40 to \$60, including the \$50 at-the-money option already solved above.

| Strike ($K$) | Market price | $d$ (moneyness, $S_0 - K$) | Implied volatility |
|---|---|---|---|
| \$40 | \$14.23 | +\$10 (in-the-money) | 40.02% |
| \$45 | \$10.57 | +\$5 | 36.00% |
| \$50 | \$7.50 | \$0 (at-the-money) | 33.25% |
| \$55 | \$4.20 | −\$5 | 26.92% |
| \$60 | \$2.19 | −\$10 (out-of-the-money) | 24.00% |

Table 5.5: Implied volatilities solved from five one-year call option market prices on the same underlying ($S_0 = \$50$, $r = 4\%$)

The pattern in Table 5.5 is the equity skew described above, not a symmetric smile: implied volatility falls by roughly 16 percentage points, from 40.02% down to 24.00%, as the strike rises from \$40 to \$60, with the largest drop, more than six points, between the \$50 and \$55 strikes. The market is pricing more downside risk into this stock than a constant-volatility price would imply: the far-out-of-the-money \$60 call is priced as if volatility were 24%, while the deep-in-the-money \$40 call, equivalent by put-call parity to a far-out-of-the-money put, is priced as if it were 40%, sixteen points higher.

Table 5.5 fixes maturity at one year and varies only the strike, one slice through what is in general a two-dimensional object. Repeating the same exercise with $T = 0.25$ (three months) instead of one year, holding $S_0$, $r$, and the same five strikes fixed, produces a second slice:

$$K = \$40 : 47.03\%, \qquad K = \$45 : 40.04\%, \qquad K = \$50 : 33.55\%,$$
$$K = \$55 : 27.04\%, \qquad K = \$60 : 24.48\%.$$

Figure 5.3 plots both slices. Both curves slope downward, so the skew is not specific to one maturity. More informative, they are not parallel: the three-month curve sits well above the one-year curve at low strikes (47.03% versus 40.02% at $K = \$40$, a gap of nearly seven points) but the two nearly converge at high strikes (24.48% versus 24.00% at $K = \$60$, under half a point apart). The three-month skew is steeper, a well-documented pattern: near-term crash protection carries a particularly large premium, since a crash is a short, sudden event, while longer-dated options leave more time for volatility to mean-revert, flattening the skew. This joint variation across strike and maturity is what the **volatility surface** describes: a grid of implied volatilities, one for every traded combination of strike and maturity, of which Table 5.5 and its three-month counterpart are two cross-sections.

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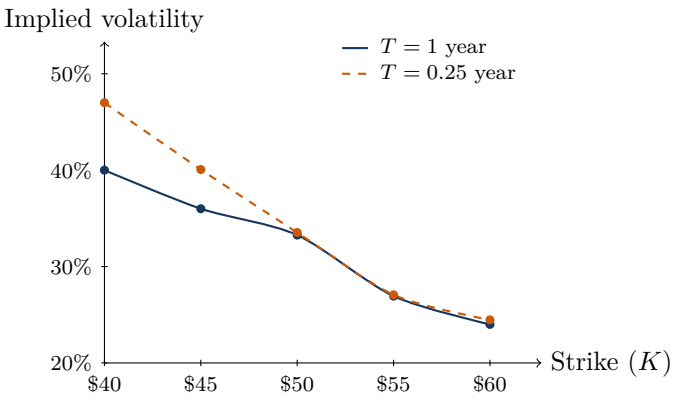



Figure 5.3: The implied volatility skew for one-year (solid) and three-month (dashed) call options on the same underlying ($S_0 = \$50$, $r = 4\%$), from Table 5.5 and its three-month counterpart

### 5.3.7 Why the Volatility Smile Matters for AI and Machine Learning

The skew and surface just computed are one of the most important places where finance meets machine learning. Black-Scholes-Merton assigns one volatility to every option on the same underlying, but, as Table 5.5 and Figure 5.3 showed, options with different strikes and maturities imply systematically different volatilities, in a pattern structured enough to be learned from data. Researchers and practitioners therefore use machine learning to model implied-volatility surfaces, to estimate the gap between theoretical and observed prices, and to build fast pricing or hedging tools for large option books.

The data are high-dimensional and highly structured: strike, maturity, underlying price, interest rates, and recent market regimes all matter. A model can learn which combinations predict unusually high or low implied volatility, but it must respect economic constraints: the fitted surface should be free of arbitrage and stable out of sample rather than memorizing recent moves. Machine learning is best viewed here as a tool for capturing patterns in the surface, not as a replacement for the no-arbitrage logic that makes option pricing coherent.

A single option's implied volatility can be distorted by events at one company, and no single option summarizes the market's volatility expectations. The CBOE Volatility Index (VIX), often called the market's "fear gauge," aggregates implied volatilities across a wide range of strikes and maturities on S&P 500 index options into one number, with a model-free methodology (following Britten-Jones and Neuberger 2000) that never inverts the Black-Scholes-Merton formula: the volatility surface condensed into one tradable index. Quoted as an annualized percentage covering the next 30 days, VIX typically trades between 12% and 20% in calm markets. Tradable through futures, options, and exchange-traded products, it recurs in this book as a market-wide risk signal: a regime filter for trading strategies in Chapter 13, and a price-based complement to the text-based sentiment measures of Chapter 16.

Figure 5.4 shows VIX's daily closing history from its 1990 inception through 9 July 2026, downloaded from Cboe's own single-file history of the index. (The companion notebook falls back to Yahoo Finance if that file cannot be reached; the two sources differ on a handful of trading days and small revisions but agree on every value quoted here.) The largest spikes are labeled: the 1998 collapse of Long-Term Capital Management amid the Russian sovereign default (Chapter 3; 45.74), the September 2001 terrorist attacks (43.74), the mid-2002 WorldCom bankruptcy and accounting-scandal selloff that closed out the dot-com bear market (45.08), the 2008 global financial crisis (80.86), the May 2010 Flash Crash (Chapter 3's own case study; 40.95), the August 2011 U.S. sovereign credit downgrade (48.00), the August 2015 selloff triggered by a surprise Chinese currency devaluation (40.74), the February 2018 "Volmageddon" collapse of short-volatility exchange-traded products (37.32), the March 2020 COVID-19 selloff (the all-time closing high of 82.69), the August 2024 yen carry-trade unwind, when a surprise Bank of Japan rate hike met a weak U.S. jobs report (65.73 intraday, 38.57 at the close), and the April 2025 tariff war (52.33, the third-highest peak of any VIX episode on record, behind only 2008 and 2020, although dozens of individual daily closes during those two

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crises were higher). Every spike has the same shape: VIX drifts in a narrow band, jumps to 40%-80% or more within days when a shock forces a repricing of risk, then subsides once the acute phase passes. This is the volatility clustering and mean reversion that Chapter 7's GARCH(1,1) recursion models.

Not every geopolitical shock qualifies: the June 2025 Israel-Iran conflict and the accompanying U.S. strikes on Iranian nuclear facilities left VIX only in the low twenties, a reminder that VIX prices *equity* volatility specifically, so a shock must threaten broad equity cash flows or risk appetite, not merely be geopolitically significant, to move it sharply.

The spike-and-decay pattern makes VIX a forecasting input rather than a description of the present: an unusually high VIX is informative because it is unlikely to persist. Backed out of option prices, VIX is the market's own forecast of the coming thirty days, and the standard empirical finding is that implied volatility carries information about future realized volatility beyond what past returns provide. That is why the risk models of Chapter 8 often add an implied-volatility term to their GARCH-style recursions.

Two cautions apply. VIX forecasts volatility, not direction, so it belongs in a risk or position-sizing model rather than a return forecast. And it is a biased forecast, sitting above realized volatility on average because index options embed compensation for crash risk, the same demand for downside protection that produces the skew. A model that treats VIX as unbiased will over-forecast volatility; the gap, the volatility risk premium, is a traded quantity, not a modeling error to be removed.

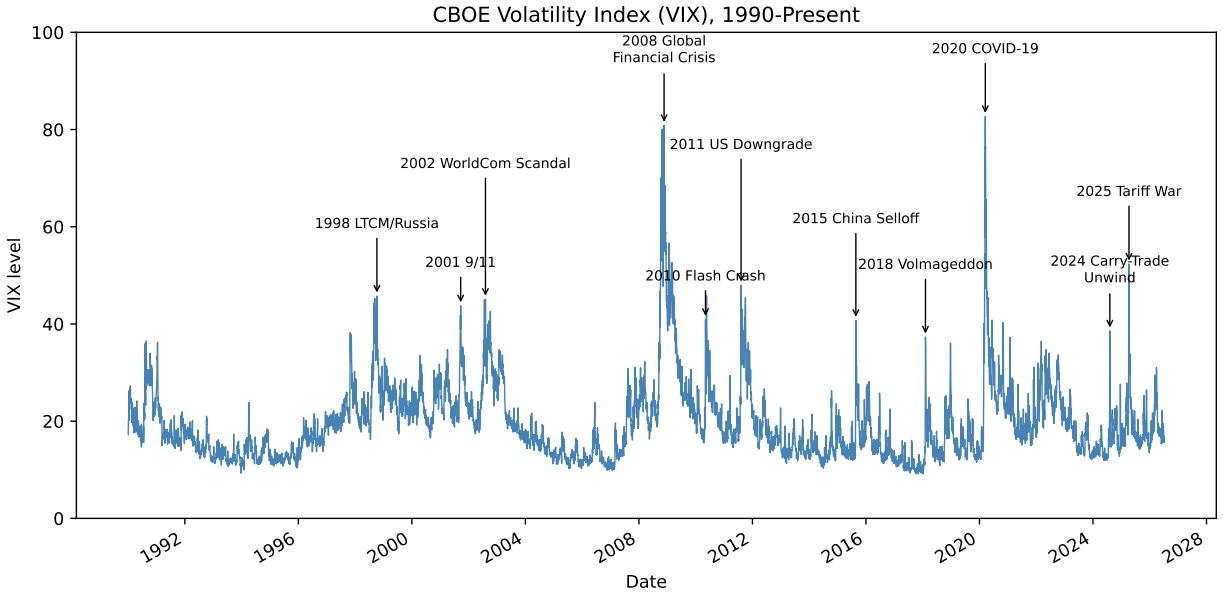



Figure 5.4: The CBOE Volatility Index (VIX), daily closes from January 1990 through 9 July 2026, from Cboe's official VIX history file. Labeled spikes correspond to the crises discussed in the text.

### 5.3.8 Real Options: Applying Option Pricing to Corporate Decisions

The option pricing machinery applies, with some adaptation, to corporate investment decisions, an application known as real options analysis. Many corporate decisions have an option-like structure: a firm has the right, but not the obligation, to take some action, and can choose whether and when to act as conditions evolve.

Consider a mining company deciding whether to develop a mineral deposit. A standard net-present-value calculation estimates a single expected value and approves the project only if it is positive. But the firm holds something closer to a call option: it can wait, watch commodity prices, and develop only if prices rise enough, walking away (losing what it spent on exploration and permitting) if they fall. This option to wait has value, just as an American option's flexibility (Section 5.3.2) has value beyond its European counterpart, and an analysis that ignores it can systematically undervalue the opportunity.

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Other common examples include the option to expand a successful pilot project, the option to abandon a failing project and salvage its assets, and the option to switch between input sources or output products as relative prices change. In each case the underlying "asset" is the value of the cash flows the decision affects, the "strike price" is the cost of acting, and volatility plays the same role as in Black-Scholes-Merton: more uncertainty makes the option to wait, expand, or abandon more valuable, not less, since flexibility is most useful when the future is most uncertain. This runs against the naive view of uncertainty as purely a negative, and it is one of the most important lessons real options analysis offers corporate finance. The qualitative logic holds even when, as is common, the underlying does not trade and no precise Black-Scholes-Merton-style number is available.

A simple numerical illustration makes this concrete. Suppose the deposit costs $K = \$50$ million to develop and, if developed today, is worth exactly \$50 million, a marginal project with NPV $= 50.0 - 50.0 = \$0$. If the firm waits one year, the deposit's value will have moved to either \$70 million (if commodity prices rise) or \$35 million (if they fall), each with probability 0.5, and the firm develops the deposit only if doing so is still profitable at that point. The expected payoff from waiting is

$$\mathbb{E}[\text{Payoff}] = 0.5 \times \max(70.0 - 50.0,\ 0) + 0.5 \times \max(35.0 - 50.0,\ 0) = 0.5 \times 20.0 + 0.5 \times 0 = \$10.0 \text{ million},$$

which, discounted one year at a risk-free rate of 5%, has a present value of $10.0/1.05 \approx \$9.52$ million, substantially higher than the \$0 NPV of developing now. The probability 0.5 is not an arbitrary guess, and discounting at the risk-free rate is legitimate only because of it: it is the risk-neutral probability of Section 5.3.1 implied by the deposit's current value, $q = (1.05 \times 50 - 35)/(70 - 35) = 0.5$, which makes today's \$50 million equal the discounted risk-neutral expectation of the two end-of-year outcomes. A real-world probability of 0.5 would instead call for a risk-adjusted discount rate, and changing either end-of-year value while holding today's value fixed changes $q$ as well (Exercise 5.17). Figure 5.5 shows the comparison as a one-step decision tree.

The value of waiting comes entirely from the option to walk away: the firm captures the upside if prices rise and avoids the downside if they fall. A point-estimate NPV, which implicitly commits to developing whatever prices do, cannot capture this asymmetry. A team that evaluates a marginal project by expected NPV alone, without asking whether waiting for information is a valuable choice, can pass over projects or approve them prematurely. Chapter 19 revisits the related American-option early-exercise decision of Section 5.3.2 with a model-free reinforcement-learning algorithm, Q-learning, which recovers the same optimal exercise policy without being told the tree's transition probabilities or payoffs.

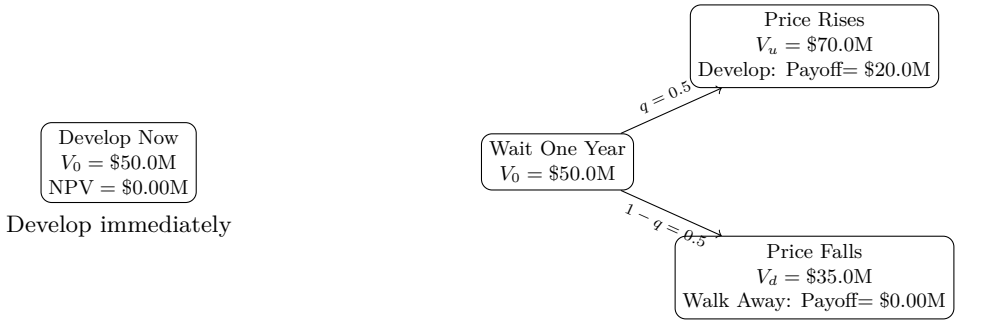



Figure 5.5: A one-step real-options decision tree for the mining deposit example: developing immediately (left) versus waiting one year to observe the commodity-price state (right)

## 5.4 Putting It Together: Valuing Meridian Fabrication

The chapter has now developed both major approaches to valuing a business, discounted cash flow and multiples. Applying both to Meridian Fabrication, with the WACC computed earlier, shows how differently they can answer the same question.

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For the DCF approach, the valuation date is the end of Year 2, the date of Meridian's latest balance sheet, so Year 3's free cash flow arrives one period from now and Year 5's three periods from now (Figure 5.6). Chapter 4's percent-of-sales forecast supplies the first of these: free cash flow recovers from −\$8.0 million in Year 2 to \$0.52 million in Year 3, as working capital absorbs less cash. Beyond that one-year forecast, suppose the recovery continues as Meridian's working-capital investment matures, to \$3.0 million in Year 4 and \$6.0 million in Year 5, after which free cash flow grows at a stable long-run rate of $g = 3\%$ forever. Discounting at the full-precision WACC computed earlier in this chapter, $9.048/112 = 8.0786\%$ (displayed there as 8.08%),

$$PV(\text{Years 3-5 FCF}) = \frac{0.52}{1.080786} + \frac{3.0}{1.080786^2} + \frac{6.0}{1.080786^3} \approx \$7.80 \text{ million},$$

and the terminal value at the end of Year 5, using the Gordon growth formula on Year 6's free cash flow, is

$$TV_5 = \frac{6.0(1.03)}{0.080786 - 0.03} \approx \$121.69 \text{ million}, \qquad PV(TV_5) = \frac{121.69}{1.080786^3} \approx \$96.39 \text{ million},$$

discounted three periods, not five, because $TV_5$ sits at the end of Year 5, exactly as far from the valuation date as Year 5's own free cash flow. The DCF enterprise value is $7.80 + 96.39 \approx \$104.19$ million, and subtracting net debt of $\$58.0 - \$9.0 = \$49.0$ million gives an implied equity value of approximately \$55.19 million. (Discounting at the rounded 8.08% instead would give \$55.16 million: the terminal value, which divides by WACC $- g$, amplifies even a rounding-level change in the discount rate.)

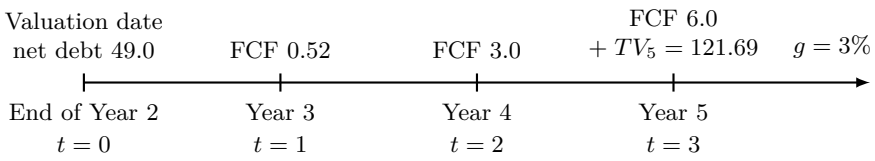



Figure 5.6: Cash-flow timeline for the Meridian DCF (\$ millions). The valuation date, $t = 0$, is the end of Year 2, so Year 3's free cash flow is discounted one period, and Year 5's free cash flow and the terminal value $TV_5$ three periods.

Compare this to the multiples-based equity value computed in Section 5.2.3: \$107.0 million, nearly twice as large. Both methods are legitimate, and the gap reflects an economic disagreement, not a computational error. The DCF value is driven by Meridian's currently weak cash generation and slow projected recovery; the multiples value assumes Meridian deserves the EBITDA multiple of healthier peers, in effect assuming its cash-flow problems are temporary.

The two methods are not even fully independent. The WACC used book equity of \$54.0 million as $E$, which the formula permits only when no market value is available; substituting the multiples-implied \$107.0 million would raise the equity weight from 48.2% to 64.8% and the WACC from 8.08% to 9.08%, which would lower the DCF value further and widen the gap. The choice of capital-structure weights is therefore itself a judgment about which of the two values is closer to the truth.

The DCF-implied equity value of \$55.19 million is close to Meridian's \$54.0 million book value of equity, while the multiples-based value is far above it. That proximity should not be over-read, since book value is a historical-cost figure and nothing forces a DCF to land near it, but it is consistent with the DCF assumptions being the more conservative starting point until there is concrete evidence, such as an improving cash conversion cycle or margin expansion, that Meridian's operating performance is converging toward that of its healthier peers.

Triangulating across methods, and explaining why they disagree, is standard practice and more useful than any single number. The next step would be to test how each estimate changes under alternative multiples, discount rates, and cash-flow scenarios; that sensitivity analysis is often more informative than the point estimate.

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## 5.5 Challenges in Valuation and Asset Pricing

The models in this chapter are elegant and, within their assumptions, exact. Applying them to real markets introduces several recurring challenges.

**Discount-rate uncertainty.** Every model in this chapter takes $r$ as given, but in practice the discount rate itself must be estimated, typically from an asset pricing model such as the CAPM or a multi-factor model (covered in Chapter 12), and estimates of the required return can vary substantially across reasonable methodologies. Since the Gordon growth model shows that price is extremely sensitive to $r - g$, uncertainty in $r$ translates directly into wide uncertainty in fair value.

**Volatility is not observed, only estimated.** The Black-Scholes-Merton formula requires $\sigma$, the volatility of the underlying asset, but $\sigma$ cannot be observed directly; it must be estimated from historical returns or backed out from other option prices as an "implied volatility." Different estimation windows and methods can produce meaningfully different prices for the same option, and volatility changes over time, the motivation for the GARCH-family models introduced in Chapter 7.

**Model risk and the limits of closed-form pricing.** Every closed-form pricing model rests on simplifying assumptions, constant volatility and continuous trading for Black-Scholes-Merton, a flat and unchanging growth rate for the Gordon growth model, no default risk in the basic bond pricing formula. Real markets violate all of these to varying degrees, and a model that is used outside the range of conditions it was built for can silently produce a confidently wrong price, a risk usually called model risk.

**Credit risk complicates bond pricing.** The bond pricing formula in this chapter assumes the promised cash flows are paid with certainty. A corporate bond carries default risk, so its price reflects both discounting and the probability-weighted possibility of missed payments, which requires the credit models developed in Chapter 9.

**Calibration versus estimation at scale.** A trading desk must reprice large portfolios of derivatives continuously as market conditions change, which requires the underlying models to be calibrated quickly and consistently across thousands of instruments. This operational requirement, not just theoretical accuracy, is a major reason simpler closed-form models such as Black-Scholes-Merton remain in wide use even though more elaborate stochastic volatility and jump-diffusion models are known to fit historical data better.

**Where machine learning fits in.** Closed-form models deviate from market prices in systematic ways (the volatility smile is a symptom), so machine learning is used both to learn the deviation (the gap between model and market price) and to learn pricing functions or implied volatility surfaces end to end. These approaches capture patterns a fixed formula cannot, but they inherit the overfitting and non-stationarity concerns of Chapter 6: a pricing model trained on historical option data can fail when conditions shift away from the training regime, usually when accurate pricing matters most.

**The term structure adds a further layer of estimation risk.** Section 5.1.4's yield curve discussion showed that a single discount rate is itself a simplification; a full analysis requires an entire curve of spot rates, and that curve must itself be estimated (typically by "bootstrapping" it from a set of observed bond prices), which introduces additional model choices and potential errors before any single bond or derivative is even priced.

**Real options are hard to value precisely.** The real options perspective of Section 5.3.8 is conceptually powerful but hard to apply with a financial option's precision, since the underlying (the value of a future business opportunity) rarely has an observable price or a clean volatility estimate. Real options analysis is therefore often used qualitatively, to check whether a standard net-present-value calculation misses significant flexibility value, rather than to produce a definitive price.

**Relative valuation inherits the market's own mistakes.** The multiples-based approach in Section 5.2.3 depends on comparable companies being reasonably priced. If a whole sector is in a speculative bubble or an unwarranted sell-off, a multiples valuation reproduces that mispricing for the target, since it prices an asset relative to its peers, not to a first-principles estimate of intrinsic value. That is why the DCF-versus-multiples comparison of Section 5.4 is informative, especially when the two disagree.

**Illiquidity complicates every valuation approach.** All of the methods in this chapter implicitly assume that once a fair value is computed, an investor could transact at or near that value. For a private company, a thinly traded bond, or a large block of stock relative to average trading volume, this assumption can fail badly: the price realizable in a sale may differ meaningfully from any theoretical value, because there are too few willing counterparties, echoing the liquidity and market-impact themes of Chapter 3.

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## 5.6 Key Takeaways

Every valuation problem in this chapter reduces to one idea: discount future cash flows, or replicate a future payoff, at a rate or under a probability measure consistent with no arbitrage. Bond pricing, the dividend discount model, and option pricing apply the same present-value logic to progressively more complex cash flows: a fixed coupon schedule, a growing perpetuity of dividends, a payoff that depends nonlinearly on another asset's price. The worked examples showed how to compute these prices and where the models break down; a practitioner who can compute a textbook price without understanding its assumptions is exposed to model risk.

The chapter also tied valuation to the statements and institutions covered earlier. The WACC example in Section 5.2.1 showed how Meridian Fabrication's balance sheet from Chapter 4 feeds into the rate used to value it. The credit-risk-adjusted bond price in Section 5.1.2 previewed the probability of default and recovery rate that Chapter 9's models estimate rather than assume. And the Greeks and implied volatility showed that even Black-Scholes-Merton is best understood as a lens for organizing risk exposures and market expectations, not a black box that outputs a single correct price. Treating elegant models as organizing frameworks, not oracles, is the mindset the rest of this book tries to instill.

The closing Meridian valuation in Section 5.4 may be the chapter's most important exercise, because it did not produce a single answer. A DCF value of \$55.19 million and a multiples-based value of \$107.0 million are both defensible and correctly computed, yet they disagree by nearly a factor of two.

The disagreement is itself informative: it isolates the assumption driving the difference, Meridian's projected cash-flow recovery versus the market's willingness to price it like a healthier peer, a more useful diagnostic than either number alone. It is also the situation a machine learning model trained to predict "fair value" would face silently: without an explicit decomposition like this one, a model blending both approaches could produce one confident-looking number while hiding an economically meaningful disagreement about which valuation philosophy fits this company now. Preserving that kind of interpretability is a theme of Chapter 18.

## 5.7 Suggested Exercises

The exercises draw on every major tool in this chapter, and several build on the Meridian Fabrication and Solstice Robotics examples from Chapter 4, so your answers can be read alongside that earlier analysis.

5.1 A 2-year bond with face value \$1,000 and an annual coupon rate of 6% is priced at a market yield of 8%. Compute its price using the discounting formula from Section 5.1.2.

5.2 Using the Gordon growth model, find the fair price of a stock expected to pay a dividend of \$2.50 next year, assuming a required return of 10% and a long-run dividend growth rate of 6%. Then recompute the price assuming growth of 8% instead, and comment on how sensitive the price is to this two-percentage-point change.

5.3 Using the one-step binomial model with $S_0 = \$40$, $u = 1.25$, $d = 0.8$, $R = 1.05$, and strike $K = \$40$, compute the risk-neutral probability $q$ and the fair price of a call option today.

5.4 Verify put-call parity using the Black-Scholes-Merton call and put prices computed in Section 5.3.3, and explain in your own words why put-call parity must hold regardless of which option pricing model is used.

5.5 Using the two-stage dividend discount model from Section 5.2.2, compute the fair price of a stock with $D_0 = \$2.00$, a high-growth rate of $g_1 = 15\%$ for the first 4 years, a terminal growth rate of $g_2 = 5\%$, and a required return of $r = 11\%$. What fraction of the total price comes from the terminal value?

5.6 A 3-year bond with a Macaulay duration of 2.86 years is priced at \$947.51 when $y = 7\%$. Use the modified-duration approximation to estimate its price if the yield falls to 6%, then explain, using the concept of convexity, whether you would expect the approximation to overstate or understate the exact repriced value.

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5.7 Explain why the Gordon growth model is more sensitive to estimation error than the bond pricing formula in this chapter, referencing the discount-rate uncertainty challenge in Section 5.5.

5.8 Describe one concrete way a Black-Scholes-Merton-based pricing system could fail silently (produce a confident but wrong price) if market conditions moved outside the model's assumptions.

5.9 Compute Meridian's WACC using Section 5.2.1's formulas if its equity beta were 0.9 instead of 1.2, holding all other inputs (risk-free rate, market risk premium, cost of debt, tax rate, and capital structure) fixed, and explain the direction of the change.

5.10 Using the Greek formulas in Section 5.3.4, explain why gamma is largest for an option that is close to at-the-money ($S_0 \approx K$) and smaller for options that are deep in- or out-of-the-money, without necessarily recomputing the formula numerically.

5.11 A sixth call option on the same stock (still $S_0 = \$50$, $r = 4\%$, $T = 1$), with a strike of $K = \$65$, trades in the market at \$1.00. Using the same numerical root-finding approach used to build Table 5.5, solve for its implied volatility, and explain whether the result is consistent with the skew pattern already observed across the \$40–\$60 strikes in Section 5.3.6.

5.12 Using the portfolio duration formula in Section 5.1.3, compute the duration of a bond portfolio holding 30% in a bond with duration 1.5 years, 50% in a bond with duration 4.0 years, and 20% in a bond with duration 8.0 years.

5.13 Using Table 5.2, explain in your own words why a corporate bond's yield should be even higher than the government yield at the same maturity, referencing the credit-risk-adjustment concept from Section 5.1.2.

5.14 Using the multiples method in Section 5.2.3 and Solstice Robotics' Chapter 4 figures (EBIT of \$27.0 million, depreciation and amortization of \$5.0 million, total debt of \$10.0 million, and cash of \$40.0 million), compute its implied enterprise and equity value assuming comparable technology firms trade at an EV/EBITDA multiple of 12×. Compare the implied equity value to its book equity of \$75.0 million.

5.15 Explain, using the DCF-versus-multiples comparison in Section 5.4, one reason a DCF valuation and a multiples-based valuation of the same company might disagree substantially, and describe what an analyst should do when they do.

5.16 Describe a real option (using the framework in Section 5.3.8) that a pharmaceutical company holds when deciding whether to continue funding a drug at each stage of clinical trials, and explain why higher uncertainty about the drug's eventual efficacy makes this option more, not less, valuable.

5.17 Using Section 5.3.8's option-to-wait example, recompute the value of waiting one year if the deposit's up-state value were \$90 million instead of \$70 million (down-state value, development cost, today's \$50 million value, and the risk-free rate unchanged). Do it twice: first holding the probability at 0.5, then using the risk-neutral probability $q$ implied by the new up-state value. Show that the first calculation raises the option's present value by less than half of the \$20 million increase in the up-state value, and explain why only the second calculation is consistent with no arbitrage.

5.18 Using Section 5.4's Meridian DCF as a template, recompute the enterprise and equity value if the terminal growth rate were $g = 2\%$ instead of 3%, holding the WACC and the Year 3-5 free cash flow forecasts fixed, and explain why the terminal growth assumption has such a large effect on the total valuation.

5.19 A junior analyst argues that because the DCF and multiples valuations for Meridian disagree so much, one of the two methods must simply be "wrong" and should be discarded. Write a short paragraph, drawing on Section 5.4, explaining why this conclusion does not necessarily follow.

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5.20 Using Table 5.2, compute the implied 1-year forward rate spanning years 4 to 5 using the 5-year and (assumed) 4-year spot rates of 4.8% and 4.7% respectively, and compare it to the two forward rates already computed in the text.

5.21 Using this chapter's delta-hedging example, recompute the hedge's net P&L if the stock instead falls from \$50 to \$49 (holding the number of shares hedged and all other option inputs fixed), and explain why the sign of the residual hedging error is the same as in the original up-move example even though the stock moved in the opposite direction.

5.22 **Challenge (real data).** Trace a real implied-volatility skew, as Section 5.3.6 did for hypothetical prices. Option chains in `yfinance` are available only for the current date, not historically, so record the date and time of your download and expect your numbers to differ from anyone else's. (a) For the S&P 500 ETF SPY, download the spot price, the current 13-week Treasury bill yield (ticker `^IRX`) as the risk-free rate, and the option chain (`Ticker("SPY").option_chain(expiry)`) for the first expiry at least 30 days away. (b) For strikes from about 85% to 110% of spot, take the bid-ask midpoint of the out-of-the-money option at each strike, converting each put to the equivalent call price with put-call parity, Equation (5.10), and invert the Black-Scholes-Merton formula, Equation (5.9), numerically for $\hat{\sigma}$. (c) Plot $\hat{\sigma}$ against moneyness $K/S_0$, and say whether you see a skew or a smile. (d) Name two simplifications in this procedure (the chapter's formula ignores SPY's dividends, and SPY options are American) and the direction in which each could bias your implied volatilities, and explain what you would do with quotes whose bid is zero, as happens outside trading hours.

## 5.8 Further Reading

- Hull (2022), *Options, Futures, and Other Derivatives*. The standard reference for derivatives pricing, covering the binomial model, Black-Scholes-Merton, and the Greeks in far greater depth and generality than this chapter.
- McDonald (2012), *Derivatives Markets*. A complementary treatment of options and forwards with a strong emphasis on the no-arbitrage replication arguments introduced in this chapter.
- Damodaran (2012), *Investment Valuation*. An extensive treatment of both discounted cash flow and relative valuation, including practical guidance on estimating discount rates, growth rates, and comparable-company multiples.
- Bodie et al. (2024), *Investments*. Covers the CAPM and cost-of-capital material in this chapter within the broader context of portfolio theory and asset pricing, previewing Chapter 12.
- Fabozzi (2007), *Fixed Income Analysis*. A thorough treatment of bond pricing, duration, convexity, and the term structure of interest rates for readers who want to go well beyond this chapter's introductory treatment.
- Brealey et al. (2025), *Principles of Corporate Finance*. Covers real options and the cost-of-capital material from a corporate finance perspective, complementing the more market-oriented treatment in this chapter.
- Black (1976), "Studies of Stock Price Volatility Changes." The original leverage-effect explanation for the equity volatility skew discussed in this chapter's implied-volatility section.
- Rubinstein (1994), "Implied Binomial Trees." The "crashophobia" explanation for the post-1987 equity skew discussed in this chapter's implied-volatility section.
- Britten-Jones and Neuberger (2000), "Option Prices, Implied Price Processes, and Stochastic Volatility." The model-free methodology underlying the VIX index discussed in this chapter's volatility-surface section.

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